

Communication Systems Lab 1

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2023AAPS0013P

August 5, 2025

1 Introduction

The tasks for this week were done in MATLAB R2022b.

2 Question 1

Question: Generate a rectangular waveform using the fourier series expansion of a rectangular wave at $f_0 = 50Hz$.

$$m(t) = \frac{4}{\pi} \sum_{n=1,3,5\dots}^N \frac{1}{n} \sin(2\pi n f_0 t) \quad (1)$$

Plot the time domain $m(t)$ and frequency domain $|M(f)|$ for different N .

Answer:

3 Question 2

Question: Generate a periodic rectangular pulse of period 20ms using the built-in square function and plot its waveform in the time domain and frequency spectrum using fft. Then, pass the rectangular pulse through a low-pass filter with impulse response $2B\text{sinc}(2Bt)$ and observe the output in the frequency domain for different values of B . Use the convolution operation.